

实验三 SQL查询

2023级图灵实验班 张昕跃 2023202300

1. 实验环境（计算机配置，操作系统，编程语言，编程工具等）

- 计算机配置: MacBook Air, Apple M3
- 操作系统: macOS Sequoia 15.1 (Build 24B83)
- 数据库管理系统 (DBMS): PostgreSQL 16.1
- 实验数据: TPC-H 数据集 (Scale Factor = 0.01)

2. 实验结果及分析

2.1 单表查询

这里对于比较简单的题目就不写过程了

1. 查询PART表中所有品牌名称以"BRAND"开头且类型包含"POLISHED"的商品，按商品尺寸降序排列。

- SQL:

```
SELECT *
FROM part
WHERE p_brand LIKE 'Brand%' AND p_type LIKE '%POLISHED%'
ORDER BY p_size DESC;
```

这里原题目似乎有笔误，因为数据库里都是以Brand开头的，以BRAND全大写什么也搜索不出来

- 运行截图:

p_partkey	p_name	p_mfgr	p_brand	p_type	p_size	p_container	p_retailprice	p_comment
1023	grey blanchd mint firebrick snow	Manufacturer#2	Brand#24	PROMO POLISHED COPPER	50	SM CASE	924.02	ideas use slyly. care
1583	cream deep floral grey chiffon	Manufacturer#1	Brand#12	ECONOMY POLISHED STEEL	50	SM JAR	1484.58	cajole fluffily
436	turquoise yellow dim purple antique	Manufacturer#1	Brand#14	LARGE POLISHED BRASS	50	WRAP CASE	1336.43	the regul
1979	tomato forest pale moccasin magenta	Manufacturer#5	Brand#52	SMALL POLISHED COPPER	50	SM PKG	1880.97	es. blithel
843	frosted metallic mint lawn blanchd	Manufacturer#1	Brand#15	MEDIUM POLISHED STEEL	50	LG PACK	1743.84	the
767	blush firebrick misty blanchd purple	Manufacturer#2	Brand#24	LARGE POLISHED TIN	50	MED DRUM	1667.76	ts. carefully unu
1333	red sky dim light ivory	Manufacturer#2	Brand#25	MEDIUM POLISHED STEEL	50	WRAP JAR	1234.33	ts alongside of the
1572	lemon midnight black papaya peru	Manufacturer#4	Brand#45	ECONOMY POLISHED COPPER	50	LG JAR	1473.57	riously ironic pinto
1753	beige smoke slate steel lavender	Manufacturer#2	Brand#24	ECONOMY POLISHED TIN	50	MED PACK	1654.75	y ironic bra
90	hot rosy violet plum pale	Manufacturer#5	Brand#51	ECONOMY POLISHED STEEL	49	JUMBO CAN	990.09	caref
97	coral dodger beige black chartreuse	Manufacturer#3	Brand#33	MEDIUM POLISHED BRASS	49	WRAP CAN	997.09	ss excuses sleep am
1863	powder misty linen wheat coral	Manufacturer#5	Brand#54	ECONOMY POLISHED BRASS	49	JUMBO BOX	1764.86	. slyly final foxes ca
635	ghost aquamarine gainsboro lemon peach	Manufacturer#5	Brand#51	PROMO POLISHED STEEL	49	WRAP JAR	1535.63	y ironic excuses.
1670	black tan metallic dark misty	Manufacturer#1	Brand#14	MEDIUM POLISHED TIN	49	WRAP CASE	1571.67	ess packages sleep, bo
1602	mint red pale midnight orange	Manufacturer#5	Brand#51	SMALL POLISHED STEEL	49	JUMBO PACK	1503.60	special theodolites
940	violet coral blanchd orange light	Manufacturer#1	Brand#15	PROMO POLISHED TIN	49	JUMBO JAR	1840.94	ate blithely bold, si
1399	brown lavender spring chocolate firebrick	Manufacturer#5	Brand#54	ECONOMY POLISHED COPPER	48	JUMBO JAR	1300.39	furiously agai
964	sky orange indian pale burlywood	Manufacturer#5	Brand#54	SMALL POLISHED BRASS	48	MED BOX	1864.96	nal asymptotes, regul
1847	yellow white puff orange rosy	Manufacturer#3	Brand#33	SMALL POLISHED COPPER	48	WRAP BOX	1748.84	ross the pending r
37	royal coral orange burnished navajo	Manufacturer#4	Brand#45	LARGE POLISHED TIN	48	JUMBO BOX	937.03	silent
20	ivory navy honeydew sandy midnight	Manufacturer#1	Brand#12	LARGE POLISHED NICKEL	48	MED BAG	920.02	are across the asympt
1616	moccasin tomato firebrick peru cyan	Manufacturer#3	Brand#33	MEDIUM POLISHED TIN	48	SM CAN	1517.61	he fl
662	lawn khaki green cornflower sienna	Manufacturer#5	Brand#53	LARGE POLISHED NICKEL	48	SM BOX	1582.68	ackages.
152	white sky antique tomato chartreuse	Manufacturer#5	Brand#53	MEDIUM POLISHED STEEL	48	MED CASE	1052.15	thely regular t
687	frosted cornsilk tomato burnished smoke	Manufacturer#5	Brand#51	LARGE POLISHED NICKEL	47	JUMBO BOX	1587.68	packages, even,
1494	blue violet white papaya royal	Manufacturer#1	Brand#15	ECONOMY POLISHED TIN	47	WRAP JAR	1395.49	of the f
750	puff slate magenta powder lawn	Manufacturer#4	Brand#43	LARGE POLISHED NICKEL	47	MED PACK	1650.75	tipliers about the q
160	frosted cornflower khaki salmon metallic	Manufacturer#5	Brand#55	STANDARD POLISHED COPPER	47	JUMBO CAN	1060.16	nts are carefully
1323	burnished firebrick light steel lemon	Manufacturer#4	Brand#45	SMALL POLISHED TIN	47	LG JAR	1224.32	y carefully
1260	tomato rosy beige yellow tan	Manufacturer#3	Brand#31	ECONOMY POLISHED STEEL	47	LG PKG	1169.26	s might
274	cyan lawn snow sky peru	Manufacturer#2	Brand#23	MEDIUM POLISHED STEEL	46	LG JAR	1174.27	hless accounts prin
224	drab lavender moccasin almond purple	Manufacturer#1	Brand#13	STANDARD POLISHED NICKEL	46	WRAP BOX	1124.22	y special depos

2. 统计CUSTOMER表中每个市场区域(MKTSEGMENT)的客户数量。

- SQL:

```
SELECT c_mktsegment, COUNT(*) AS customer_count
FROM customer
GROUP BY c_mktsegment;
```

- 运行截图:

```
tpch_db=# SELECT c_mktsegment, COUNT(*) AS customer_count
FROM customer
GROUP BY c_mktsegment;
 c_mktsegment | customer_count
-----+-----
 MACHINERY    |             288
 AUTOMOBILE   |             302
 BUILDING     |             337
 FURNITURE    |             279
 HOUSEHOLD    |             294
(5 行记录)
```

3. 查询SUPPLIER表中，账户余额(ACCTBAL)平均值大于9000的国家(NATIONKEY)及其平均余额。

- SQL:

```
SELECT s_nationkey, AVG(s_acctbal) AS avg_balance
FROM supplier
GROUP BY s_nationkey
HAVING AVG(s_acctbal) > 9000;
```

由于没有账户余额均值大于9000，所以这里是空的

- 运行截图:

```
tpch_db=# SELECT s_nationkey, AVG(s_acctbal) AS avg_balance
FROM supplier
GROUP BY s_nationkey
HAVING AVG(s_acctbal) > 9000;
 s_nationkey | avg_balance
-----+-----
(0 行记录)
```

4. 查询ORDERS表中订单总金额(TOTALPRICE)最高的前10笔订单信息。

- SQL:

```
SELECT *
FROM orders
ORDER BY o_totalprice DESC
LIMIT 10;
```

- 运行截图：

```
tpch_db=# SELECT *
FROM orders
ORDER BY o_totalprice DESC
LIMIT 10;
```

o_orderkey	o_custkey	o_orderstatus	o_totalprice	o_orderdate	o_orderpriority	o_clerk	o_shippriority	o_comment
52965	676	O	466001.28	1996-09-22	3-MEDIUM	Clerk#000000103	0	thless pinto beans use about the blithe
29158	667	O	439687.23	1995-10-21	2-HIGH	Clerk#000000525	0	lessly quiet deposits across the furiously ironi
44707	1013	O	431771.98	1997-08-14	1-URGENT	Clerk#000000313	0	ly fluffily special ideas.
59106	953	O	430619.75	1996-10-24	4-NOT SPECIFIED	Clerk#000000344	0	y above the slyly final deposits, furiously regular pinto beans use s
6882	178	O	422359.65	1997-04-09	1-URGENT	Clerk#000000974	0	y dolphins, furiously ironic asymptotes s
57376	676	O	411255.46	1995-06-20	1-URGENT	Clerk#000000618	0	ular Tiresias after the furiously regular requests sha
39456	1016	O	409770.83	1998-02-16	2-HIGH	Clerk#000000972	0	ts sleep final requests, carefully b
17571	358	F	408345.74	1992-03-16	4-NOT SPECIFIED	Clerk#000000503	0	ckages, blithely silent requests use quickly final packages, somas mainta
38620	1279	F	406938.36	1994-10-05	2-HIGH	Clerk#000000457	0	nic packages sleep quickly regular excuses, busy packages nag finally! package
35460	334	F	405742.27	1993-11-28	5-LOW	Clerk#000000112	0	deposits wake furiously into the s

(10 行记录)

5. 查询LINEITEM表中每条订单明细的实际金额（扩展价格×(1-折扣)），并显示订单号、商品号和实际金额。

- SQL：

```
SELECT
    l_orderkey,
    l_partkey,
    l_extendedprice * (1 - l_discount) AS actual_price
FROM lineitem;
```

- 运行截图：

l_orderkey	l_partkey	actual_price
1	1552	23721.9360
1	674	51586.1892
1	637	11070.9360
1	22	23493.0696
1	241	24650.7840
1	157	31460.7840
2	1062	36596.2800
3	43	39890.5920
3	191	48121.4790
3	1285	30107.7864
3	294	2364.6942
3	1831	46578.4704
3	622	35629.3080
4	881	51852.7080
5	1086	14510.0760
5	1240	27505.1822

5		1240		27595.1832
5		376		58713.0200
6		1397		44197.1956
7		1821		19226.6712
7		1453		11214.8460
7		948		76546.1160
7		1631		41626.2308
7		1519		49661.0296
7		793		55725.6910
7		1573		7077.9360
32		828		45986.6120
32		1980		59018.8928
32		442		2443.2408
32		28		3377.9928
32		859		73561.7300
32		117		5858.5536
33		614		42727.1481
33		606		47246.9760
33		1375		6062.7575
33		340		46277.0854
34		884		23203.4400
34		895		36348.8136
34		1696		9394.4172
35		5		21285.6000
35		1620		48630.9752
35		1209		7305.1160
35		852		41191.9750
35		1200		34445.5360
35		308		32817.4280
36		1198		42011.0418
37		227		41030.8080

37	227	41858.8888
37	1268	43321.0830
37	130	42080.8105
38	1759	70150.0800
39	24	36997.7608
39	1866	42287.2112
39	679	68304.9308
39	206	32920.5120
39	546	61579.2078
39	944	69369.7440
64	860	35129.1570
65	597	37769.2198
65	739	36074.0600
65	14	17466.7311

6. 统计PART表中包装类型(CONTAINER)以"SM"开头的各类包装的商品数量。

- SQL:

```
SELECT p_container, COUNT(*) AS p_counts
FROM part
WHERE p_container LIKE 'SM%'
GROUP BY p_container;
```

- 运行截图:

```
tpch_db=# SELECT p_container, COUNT(*) AS p_counts
FROM part
WHERE p_container LIKE 'SM%'
GROUP BY p_container;
```

p_container	p_counts
SM PACK	61
SM CAN	50
SM CASE	44
SM DRUM	52
SM JAR	54
SM BAG	58
SM PKG	59
SM BOX	36

(8 行记录)

7. 查询LINEITEM表中，发货模式(SHIPMODE)为"AIR"或"TRUCK"，且平均数量大于25的发货模式。

- SQL:

```
SELECT l_shipmode, AVG(l_quantity) AS avg_quantity
FROM lineitem
WHERE l_shipmode IN ('AIR', 'TRUCK')
GROUP BY l_shipmode
HAVING AVG(l_quantity) > 25;
```

- 运行截图:

```
tpch_db=# SELECT l_shipmode, AVG(l_quantity) AS avg_quantity
FROM lineitem
WHERE l_shipmode IN ('AIR', 'TRUCK')
GROUP BY l_shipmode
HAVING AVG(l_quantity) > 25;
 l_shipmode |      avg_quantity
-----+-----
TRUCK      | 25.7071182548794489
AIR        | 25.4776822517960193
(2 行记录)
```

8. 查询CUSTOMER表，计算每个客户的折扣余额（账户余额×0.9），并按折扣余额降序排列。

- SQL:

```

SELECT
  c_custkey,
  c_name,
  c_acctbal,
  c_acctbal * 0.9 AS discounted_balance
FROM customer
ORDER BY discounted_balance DESC;

```

- 运行截图：

c_custkey	c_name	c_acctbal	discounted_balance
213	Customer#000000213	9987.71	8988.939
45	Customer#000000045	9983.38	8985.042
1106	Customer#000001106	9977.62	8979.858
200	Customer#000000200	9967.60	8970.840
140	Customer#000000140	9963.15	8966.835
381	Customer#000000381	9931.71	8938.539
43	Customer#000000043	9904.28	8913.852
100	Customer#000000100	9889.89	8900.901
780	Customer#000000780	9874.12	8886.708
518	Customer#000000518	9871.66	8884.494
197	Customer#000000197	9860.22	8874.198
219	Customer#000000219	9858.57	8872.713
987	Customer#000000987	9850.64	8865.576
1042	Customer#000001042	9849.87	8864.883
468	Customer#000000468	9834.19	8850.771
1370	Customer#000001370	9802.04	8821.836
1479	Customer#000001479	9793.29	8813.961
574	Customer#000000574	9787.56	8808.804
1123	Customer#000001123	9786.36	8807.724
1403	Customer#000001403	9782.34	8804.106
1204	Customer#000001204	9777.19	8799.471
1051	Customer#000001051	9776.39	8798.751
919	Customer#000000919	9774.97	8797.473
157	Customer#000000157	9768.73	8791.857
601	Customer#000000601	9768.21	8791.389
734	Customer#000000734	9750.71	8775.639
487	Customer#000000487	9749.37	8774.433
145	Customer#000000145	9748.93	8774.037
1086	Customer#000001086	9726.83	8754.147
592	Customer#000000592	9712.75	8741.475
1478	Customer#000001478	9701.54	8731.386
1026	Customer#000001026	9699.28	8729.352
663	Customer#000000663	9698.21	8728.389
1460	Customer#000001460	9680.51	8712.459
606	Customer#000000606	9676.98	8709.282
849	Customer#000000849	9670.64	8703.576
279	Customer#000000279	9663.23	8696.907
529	Customer#000000529	9647.58	8682.822

1386	Customer#000001386	9643.87	8679.483
810	Customer#000000810	9632.77	8669.493
614	Customer#000000614	9630.24	8667.216
1195	Customer#000001195	9621.49	8659.341
768	Customer#000000768	9618.84	8656.956
945	Customer#000000945	9615.39	8653.851
950	Customer#000000950	9609.77	8648.793
1022	Customer#000001022	9605.83	8645.247
1198	Customer#000001198	9593.35	8634.015
711	Customer#000000711	9591.51	8632.359
251	Customer#000000251	9585.32	8626.788
246	Customer#000000246	9584.96	8626.464
691	Customer#000000691	9566.15	8609.535
904	Customer#000000904	9562.82	8606.538
7	Customer#000000007	9561.95	8605.755
557	Customer#000000557	9559.04	8603.136
218	Customer#000000218	9541.19	8587.071
407	Customer#000000407	9537.08	8583.372
188	Customer#000000188	9533.37	8580.033
1130	Customer#000001130	9519.36	8567.424
1167	Customer#000001167	9510.87	8559.783

9. 查询SUPPLIER表中，电话号码以"15"开头的供应商，按国家分组后，筛选出供应商数量超过5个的国家。

- SQL:

```
SELECT s_nationkey, COUNT(*) AS supplier_count
FROM supplier
WHERE s_phone LIKE '15%'
GROUP BY s_nationkey
HAVING COUNT(*) > 5;
```

这里为了看看是不是真的没有，我查了发现以15开头的供应商就只有3个，因此不管怎么分都不会有超过五个的国家。

s_suppkey	s_name	s_address	s_nationkey	s_phone	s_acctbal	s_comment
2	Supplier#000000002	89eJ5ksX3ImxJ0Bvx0bC,	5	15-679-861-2259	4032.68	slyly bold instructions. idle dependen
63	Supplier#000000063	NlV00Qy1oPvPkW5AYuWgomX,hgqm1	5	15-761-401-3047	5742.03	ar deposits. blithely bold accounts against the slyly final pinto beans sleep about th
78	Supplier#000000078	9y30Z2CV hGrsrQxzB7V3zTtygHVH1G3SD6yrz	5	15-670-998-6860	1044.10	, regular packages wake quickly bold requests. carefully unusual requests about the un

- 运行截图:

```

tpch_db=# SELECT s_nationkey, COUNT(*) AS supplier_count
FROM supplier
WHERE s_phone LIKE '15%'
GROUP BY s_nationkey
HAVING COUNT(*) > 5;
 s_nationkey | supplier_count
-----+-----
(0 行记录)

```

10. 查询ORDERS表中订单优先级(PRIORITY)包含"URGENT"的订单，按订单日期升序、总金额降序排列，显示前15条。

- SQL:

```

SELECT *
FROM orders
WHERE o_orderpriority LIKE '%URGENT%'
ORDER BY o_orderdate ASC, o_totalprice DESC
LIMIT 15;

```

- 运行截图:

```

tpch_db=# SELECT *
FROM orders
WHERE o_orderpriority LIKE 'URGENT%'
ORDER BY o_orderdate ASC, o_totalprice DESC
LIMIT 15;

```

o_orderkey	o_custkey	o_orderstatus	o_totalprice	o_orderdate	o_orderpriority	o_clerk	o_shippriority	o_comment
3271	331	F	138053.06	1992-01-01	1-URGENT	Clerk#000000421	0	s. furiously regular requests
1248	472	F	295111.15	1992-01-02	1-URGENT	Clerk#000000890	0	t the carefully regular dugouts. s
36582	34	F	199609.11	1992-01-03	1-URGENT	Clerk#000000358	0	even requests might use carefully
47591	823	F	125040.01	1992-01-03	1-URGENT	Clerk#000000497	0	eas nag furiously bravely regular depths. even i
35811	586	F	185551.43	1992-01-04	1-URGENT	Clerk#000000257	0	egular deposits. special ideas hin
9379	350	F	152004.78	1992-01-04	1-URGENT	Clerk#000000764	0	nusual deposits. furiously final packages wake evenly even tithes. qu
55011	833	F	214711.38	1992-01-05	1-URGENT	Clerk#000000957	0	ke carefully regular reques
31555	679	F	105491.57	1992-01-05	1-URGENT	Clerk#000000049	0	above the furiously special packages. furiously special
18625	1489	F	283425.83	1992-01-06	1-URGENT	Clerk#000000512	0	its use furiously abou
57698	953	F	189185.83	1992-01-07	1-URGENT	Clerk#000000753	0	s try to lose slyly across the fluff
2983	613	F	83588.10	1992-01-07	1-URGENT	Clerk#000000278	0	r the even requests. accounts maintain. regular accounts
11206	832	F	378166.33	1992-01-10	1-URGENT	Clerk#000000966	0	s— ironic, unusual requests haggle furiously. carefully special depend
34662	760	F	251035.71	1992-01-11	1-URGENT	Clerk#000000878	0	xcuses wake. express. pending theodolites lose furiously blithely final req
44033	1370	F	59313.16	1992-01-11	1-URGENT	Clerk#000000109	0	requests nag quickly express requests. express pinto beans hag
47879	1165	F	54960.85	1992-01-12	1-URGENT	Clerk#000000528	0	dolites nag. regular instructions cajole fur

(15 行记录)

11. 查询PART表，找出品牌名称第二个字符为大写字母"R"，且制造商名称以"Manufacturer"开头的商品，计算每个尺寸组的平均零售价。

- SQL:

```

SELECT p_size, AVG(p_retailprice) AS avg_retailprice
FROM part
WHERE p_mfgr LIKE 'Manufacturer%' AND p_brand LIKE '_R%'
-- AND p_brand LIKE '_R%' 是无结果的
GROUP BY p_size
ORDER BY p_size;
-- 加上ORDER BY便于查看

```

这里似乎也是笔误，查询大写R就是无结果的：

```
tpch_db=# SELECT p_size, AVG(p_retailprice) AS avg_retailprice
FROM part
WHERE p_mfgr LIKE 'Manufacturer%' AND p_brand LIKE '_R%'
-- AND p_brand LIKE '_R%' 是无结果的
GROUP BY p_size
ORDER BY p_size;
 p_size | avg_retailprice
-----+-----
(0 行记录)
```

- 运行截图：

```
tpch_db=# SELECT p_size, AVG(p_retailprice) AS avg_retailprice
FROM part
WHERE p_mfgr LIKE 'Manufacturer%' AND p_brand LIKE '_r%'
-- AND p_brand LIKE '_R%' 是无结果的
GROUP BY p_size
ORDER BY p_size;
 p_size |      avg_retailprice
-----+-----
1 | 1404.1318367346938776
2 | 1386.9191666666666667
3 | 1401.5525000000000000
4 | 1417.5124324324324324
5 | 1379.8811111111111111
6 | 1490.9362162162162162
7 | 1357.6187500000000000
8 | 1404.7036734693877551
9 | 1440.9705128205128205
10 | 1388.8791666666666667
11 | 1389.9090000000000000
12 | 1402.8688571428571429
13 | 1413.2795454545454545
14 | 1368.4354054054054054
15 | 1424.9637037037037037
16 | 1348.6930000000000000
17 | 1411.3565000000000000
18 | 1368.5989189189189189
19 | 1357.7855555555555556
20 | 1413.9482000000000000
21 | 1346.3275000000000000
22 | 1408.4171428571428571
23 | 1339.4060000000000000
24 | 1375.0407142857142857
25 | 1306.8892682926829268
26 | 1460.6954000000000000
27 | 1392.3934883720930233
28 | 1387.6402631578947368
29 | 1406.1179411764705882
30 | 1420.8047368421052632
31 | 1430.3635135135135135
32 | 1282.3482857142857143
```

```
33 | 1400.0728947368421053
34 | 1473.7392682926829268
35 | 1408.0480000000000000
36 | 1372.9095348837209302
37 | 1438.8561290322580645
38 | 1519.9000000000000000
39 | 1455.0216666666666667
40 | 1434.6504878048780488
41 | 1432.5550000000000000
42 | 1355.8772222222222222
43 | 1397.4625714285714286
44 | 1456.4323809523809524
45 | 1337.4560465116279070
46 | 1299.2415151515151515
47 | 1434.4118604651162791
48 | 1461.1500000000000000
49 | 1380.6948780487804878
50 | 1480.8006451612903226
(50 行记录)
```

12. 查询LINEITEM表，统计1995-1996年间每个季度的订单明细数量，只显示数量超过1000的季度。

- SQL:

```
SELECT
    EXTRACT(YEAR FROM l_shipdate) AS year,
    EXTRACT(QUARTER FROM l_shipdate) AS quarter,
    COUNT(*) AS item_count
FROM lineitem
WHERE l_shipdate BETWEEN DATE '1995-01-01' AND DATE '1996-12-31'
GROUP BY year, quarter
HAVING COUNT(*) > 1000
ORDER BY year, quarter;
```

- 运行截图:

```

tpch_db=# SELECT
    EXTRACT(YEAR FROM l_shipdate) AS year,
    EXTRACT(QUARTER FROM l_shipdate) AS quarter,
    COUNT(*) AS item_count
FROM lineitem
WHERE l_shipdate BETWEEN DATE '1995-01-01' AND DATE '1996-12-31'
GROUP BY year, quarter
HAVING COUNT(*) > 1000
ORDER BY year, quarter;
 year | quarter | item_count
-----+-----+-----
 1995 |         | 2100
 1995 |         | 2135
 1995 |         | 2236
 1995 |         | 2302
 1996 |         | 2284
 1996 |         | 2284
 1996 |         | 2305
 1996 |         | 2327
(8 行记录)

```

2.2 多表查询

1. 查询每个国家(NATION)的客户数量，显示国家名称和客户数量。

- SQL:

```

SELECT n_name, COUNT(*) AS customer_count
from customer c, nation n
where c.c_nationkey = n.n_nationkey
GROUP BY n_name
ORDER BY customer_count DESC;

SELECT n_name, COUNT(c.c_custkey) AS customer_count
FROM nation n
JOIN customer c ON n.n_nationkey = c.c_nationkey
GROUP BY n_name
ORDER BY customer_count DESC;

```

用 where+ =拼接 和 JOIN 都可以，后面我大多用JOIN

- 运行截图:

```

tpch_db=# SELECT n_name , COUNT(*) AS customer_count
from customer c, nation n
where c.c_nationkey = n.n_nationkey
GROUP BY n_name
ORDER BY customer_count DESC;
          n_name          | customer_count
-----+-----
  IRAN                    |              72

```

MOROCCO	72
CANADA	69
BRAZIL	68
SAUDI ARABIA	67
JAPAN	67
EGYPT	66
INDONESIA	66
ROMANIA	64
MOZAMBIQUE	62
ALGERIA	61
INDIA	60
ARGENTINA	59
RUSSIA	59
VIETNAM	58
IRAQ	58
CHINA	58
ETHIOPIA	57
GERMANY	57
UNITED KINGDOM	56
PERU	56
JORDAN	54
KENYA	50
UNITED STATES	48
FRANCE	36

(25 行记录)

tpch_db=#

```
SELECT n.n_name, COUNT(c.c_custkey) AS customer_count
FROM nation n
JOIN customer c ON n.n_nationkey = c.c_nationkey
GROUP BY n.n_name
ORDER BY customer_count DESC;
```

n_name	customer_count
-----+-----	
IRAN	72
MOROCCO	72
CANADA	69
BRAZIL	68
SAUDI ARABIA	67
JAPAN	67
EGYPT	66
INDONESIA	66
ROMANIA	64
MOZAMBIQUE	62
ALGERIA	61
INDIA	60
ARGENTINA	59
RUSSIA	59
VIETNAM	58
IRAQ	58

CHINA	58
ETHIOPIA	57
GERMANY	57
UNITED KINGDOM	56
PERU	56
JORDAN	54
KENYA	50
UNITED STATES	48
FRANCE	36

(25 行记录)

2. 查询每个地区(REGION)的供应商数量，显示地区名称和供应商数量

- SQL:

```
SELECT r.r_name AS region_name, COUNT(s.s_suppkey) AS supplier_count
FROM supplier s
JOIN nation n ON n.n_nationkey = s.s_nationkey
JOIN region r ON r.r_regionkey = n.n_regionkey
GROUP BY r.r_name;
```

- 运行截图:

```
tpch_db=# SELECT r.r_name AS region_name, COUNT(s.s_suppkey) AS supplier_count
FROM supplier s
JOIN nation n ON n.n_nationkey = s.s_nationkey
JOIN region r ON r.r_regionkey = n.n_regionkey
GROUP BY r.r_name;
      region_name      | supplier_count
-----+-----
ASIA                   | 27
MIDDLE EAST           | 12
AMERICA                | 20
EUROPE                 | 20
AFRICA                 | 21
(5 行记录)
```

3. 查询单个订单总金额超过10000的客户名称及其订单总金额。

- SQL:

```
SELECT c.c_name AS customer_name, o.o_totalprice AS total_amount
FROM customer c
JOIN orders o ON c.c_custkey = o.o_custkey
WHERE o.o_totalprice > 10000
ORDER BY o.o_totalprice DESC;
```

这里其实表述不是很清楚，订单总金额指的是单个订单还是这个客户的所有订单呢，如果是所有订单就要像下面这样写了

不过我们暂且还是理解成单个

```
SELECT c.c_name AS customer_name, SUM(o.o_totalprice) AS total_amount
FROM customer c
JOIN orders o ON c.c_custkey = o.o_custkey
GROUP BY c.c_name
HAVING SUM(o.o_totalprice) > 10000
ORDER BY SUM(o.o_totalprice) DESC;
```

- 运行截图：

customer_name	total_amount
Customer#000000676	466001.28
Customer#000000667	439687.23
Customer#000001013	431771.98
Customer#000000953	430619.75
Customer#000000178	422359.65
Customer#000000676	411255.46
Customer#000001016	409770.83
Customer#000000358	408345.74
Customer#000001279	406938.36
Customer#000000334	405742.27
Customer#000001003	405401.76
Customer#000000154	405235.90
Customer#000000583	403464.01
Customer#000000166	402930.49
Customer#000000085	401055.62
Customer#000001210	400191.77
Customer#000000772	397797.80
Customer#000001303	397549.76
Customer#000001096	396261.24
Customer#000000787	395495.85
Customer#000001375	395039.05
Customer#000001426	392022.76

Customer#000001436		393923.76
Customer#000000424		392614.07
Customer#000000061		392172.90
Customer#000001147		386714.93
Customer#000000379		386334.55
Customer#000000458		384265.43
Customer#000001318		382233.07
Customer#000000665		381793.12
Customer#000000805		381170.13
Customer#000000214		379837.62
Customer#000000455		379689.83
Customer#000000832		378166.33
Customer#000000107		377636.63
Customer#000001477		376904.18
Customer#000001441		376309.96
Customer#000001453		373674.80
Customer#000001117		372954.40
Customer#000001360		372934.56
Customer#000001025		372916.06
Customer#000000545		372488.09
Customer#000001355		371804.24
Customer#000001082		371624.55
Customer#000000787		370957.42
Customer#000000955		370518.49
Customer#000000488		369844.66
Customer#000000706		369804.91
Customer#000000364		369658.21
Customer#000000457		368982.60
Customer#000000281		368220.47

```
Customer#000000538 | 367176.04
Customer#000000694 | 366949.49
Customer#000000802 | 366916.27
Customer#000000455 | 366528.14
Customer#000000260 | 366479.63
Customer#000001165 | 366277.63
Customer#000000214 | 365518.53
Customer#000001226 | 364437.75
Customer#000000737 | 362967.77
:
```

4. 查询至少有一个订单的客户信息。

- SQL:

```
SELECT c.*
FROM customer c
WHERE EXISTS (
    SELECT 1
    FROM orders o
    WHERE o.o_custkey = c.c_custkey
);

-- Use join
SELECT DISTINCT c.*
FROM customer c
JOIN orders o ON c.c_custkey = o.o_custkey
ORDER BY c.c_custkey;
```

直接思路是直接从customer表中选然后一个个判断有订单还是没有
当然也可以直接把customer表和orders表 JOIN 起来，会自动过滤掉没有订单的
不过这里每个客户可能有多个订单，因此加一个 DISTINCT 去重

- 运行截图:

c_custkey	c_name	c_address	c_nationkey	c_phone	c_acctbal	c_mktsegment	c_comment
1	Customer#000000001	IvhzIApeRb ot,c,E	15	25-989-741-2988	711.56	BUILDING	to the even, regular platelets. regular, ironic epitaphs nag e
2	Customer#000000002	XSTf4,NcWdYvMNeEtEgvvmRchLXak	13	23-768-687-3665	121.65	AUTOMOBILE	l accounts. blithely ironic theodolites integrate boldly: caref
4	Customer#000000004	XxvSjSLAGtn	4	14-128-100-5944	2866.83	MACHINERY	requests, final, regular ideas sleep final accou
5	Customer#000000005	KvpyuHCplRB4wgAiGV6sYpZqTtj	3	13-758-942-6364	794.47	HOUSEHOLD	n accounts will have to unwind. foxes cajole accor
7	Customer#000000007	TcGe5gaZngVePeXUSkRrvXBfkaSDtea	18	28-198-982-9759	9561.95	AUTOMOBILE	ainst the ironic, express theodolites. express, even pinto beans among the e
8	Customer#000000008	IOE10bB0AymnC, 0PrRYBCPlYgJ8xcBPMwhlS	17	27-147-574-9335	6819.74	BUILDING	among the slyly regular theodolites kindle blithely courts. carefully even t
10	Customer#000000010	6LrEaVKRGKPLVcg12ArL Q3rqzLzcT1 v2	5	15-741-346-9870	2753.54	HOUSEHOLD	es regular deposits haggle. fur
11	Customer#000000011	PkWs 3HlXqWtuzrk9g33BE1	23	33-464-151-3439	-272.60	BUILDING	ckages. requests sleep slyly. quickly even pinto beans promise above the sly
13	Customer#000000013	nsXQu0vJDP7PM6S9uCS3RSp	3	13-761-547-5974	3857.34	BUILDING	ounts sleep carefully after the close frays. carefully bold notornis use iro
14	Customer#000000014	KXK1etM1L210EA	1	11-845-129-3851	5266.38	FURNITURE	, ironic packages across the unus
16	Customer#000000016	cYlaeMLZSMAQ02 d0W,	10	20-781-689-3187	4681.03	FURNITURE	kly silent courts. thinly regular theodolites sleep fluffily after
17	Customer#000000017	1zrh 6jdgtp2eqdtbksW08S64S2XruMfIXyR7	2	12-978-682-3487	6.34	AUTOMOBILE	ackages wake! blithely even pint
19	Customer#000000019	uc,3bHIX84H,wdrnL0jVs1qXCqZtr	18	28-396-526-5053	8914.71	HOUSEHOLD	nag. furiously careful packages are slyly at the accounts. furiously regula
20	Customer#000000020	JrPk8PqLj4Ne	22	32-957-234-8742	7603.40	FURNITURE	g alongside of the special excuses-- fluffily enticing packages wake
22	Customer#000000022	QI6p41,FN5k7RZocCVPUTKUdYpB	3	13-806-545-9701	591.98	MACHINERY	s nod furiously above the furiously ironic ideas.
23	Customer#000000023	odY W13N7Be30C5MpgfmcYss0Wn6TKT	3	13-312-472-8245	3332.02	HOUSEHOLD	deposits. special deposits cajole slyly. fluffily special deposits about the
25	Customer#000000025	Hp8GyF0gGHFYS1LH5t8fe	12	22-603-468-3533	7133.70	FURNITURE	y. accounts sleep ruthlessly according to the regular theodolites. unusual i
26	Customer#000000026	8ljrScZEM17UciP	22	32-363-455-4837	5182.05	AUTOMOBILE	c requests use furiously ironic requests. slyly ironic dependencies us
28	Customer#000000028	iVyg0daQ,Tha8x2WPWA9m2529m	8	18-774-241-1462	1007.18	FURNITURE	along the regular deposits. furiously final pac
29	Customer#000000029	sJ5adtffyAKCK63df2,VF25zyQMVYE34uh	0	10-773-203-7342	7618.27	FURNITURE	its after the carefully final platelets x-ray against
31	Customer#000000031	UUACh00v1aAv6eXDAebryDB xlvst	23	33-197-837-7494	5236.89	HOUSEHOLD	s use among the blithely pending depo
32	Customer#000000032	j0zZ2zi UnId,DcNBkXj9u0Tlpzi06Zc03j	15	25-436-914-2184	3471.53	BUILDING	cial ideas. final, furious requests across the e
34	Customer#000000034	06G9wZ6dnczmt0x589xgE,M2KV	15	25-344-968-5422	8589.70	HOUSEHOLD	nder against the even, pending accounts. even
35	Customer#000000035	TEjWGE4nBzL2L	17	27-566-888-7431	1228.24	HOUSEHOLD	requests. special, express requests nag slyly furiously
37	Customer#000000037	7EV4Pwh,3SboctTWt	8	18-385-235-7162	-917.75	FURNITURE	ilent packages are carefully among the deposits. furiously
38	Customer#000000038	aSEeSe9S6RRRLP Zap7	12	22-306-880-7212	6345.11	HOUSEHOLD	lar excuses. closely even asymptotes cajole blithely excuses. carefully sile
40	Customer#000000040	g0nGWaY5V1ofv	3	13-652-915-8939	1335.38	BUILDING	rges impress after the slyly ironic courts. foxes are, blithely
41	Customer#000000041	IM9mzmxyoeBmw8LA7G3Ydska2nkZF	10	20-917-711-4011	278.95	HOUSEHOLD	ly regular accounts hang bold, silent packages. unusual foxes haggle slyly a
43	Customer#000000043	ouSbJHk8lh5FKX3zGso3ZSjJ9Aa3PoaFd	19	29-316-665-2897	9904.28	MACHINERY	ial requests: carefully pending foxes detect quickly. carefully final courts
44	Customer#000000044	0i,d0SPw0u4j04x,,P85E0dmhZGvMtBw1	16	26-198-260-5375	7315.94	AUTOMOBILE	r requests around the unusual, bold a
46	Customer#000000046	eaTXWm10L9	6	16-357-681-2087	5744.59	AUTOMOBILE	ctions. accounts sleep furiously even requests. regular, regular accounts ca
47	Customer#000000047	b0UgocSqEW5 gdVbnMT	2	12-427-271-9466	274.58	BUILDING	ions. express, ironic instructions sleep furiously ironic ideas. furi
49	Customer#000000049	0u0eAe7Fgrd7F9W09EyuUa4nxT,68L FXAxzl	10	20-908-631-4424	4573.94	FURNITURE	usual foxes! fluffily pending packages maintain to the regular
50	Customer#000000050	95zDYlKzrByy1l0EITl o	6	16-658-112-3221	4266.13	MACHINERY	ts, furiously ironic accounts cajole furiously slyly ironic dinos.
52	Customer#000000052	7 00qGqoS9jfv518C7ljchJSD0	11	21-186-284-5998	5638.28	HOUSEHOLD	ic platelets use evenly even accounts. stealthy theodolites cajole furiou
53	Customer#000000053	HnaxHzTFFTZs8MuCpJyTBz47Cm4wF00gib	15	25-168-852-5363	4113.64	HOUSEHOLD	ar accounts are. even foxes are blithely. fluffily pending deposits boost
55	Customer#000000055	z1RBR4KNEl HzaiV3a i9n6elrxzDEh8r8pDom	10	20-188-440-8525	4572.11	MACHINERY	ully unusual packages wake bravely bold packages. unusual requests boost dep
56	Customer#000000056	BjYZY3Qk4yD5B	10	20-895-685-6920	6538.86	FURNITURE	. notornis wake carefully. carefully fluffly requests are furiously even acco
58	Customer#000000058	g9a970k15v9fcXEWjpmYpBZIRUohi T	13	23-244-493-2508	6478.46	HOUSEHOLD	ideas. ironic ideas affix furiously express, final instructions. regular exc
59	Customer#000000059	zLCPWw920t8ihgsp0G14	1	11-355-584-3112	3458.60	MACHINERY	ously final packages haggle blithely after the express deposits. furiou
61	Customer#000000061	8knvde4EAjzhg3ver Bfx7Aq0sT39o gtqjaYE	17	27-626-559-8599	1536.24	FURNITURE	regular packages shall have to impress along the
62	Customer#000000062	upJK2Dmw13,	7	17-361-978-7059	595.61	MACHINERY	kly special dolphins. pinto beans are slyly. quickly regular accounts are fu

5. 查询每个国家的平均客户账户余额，结果显示国家名称和平均客户账户余额。

- SQL:

```
SELECT n.n_name AS nation_name, AVG(c.c_acctbal) AS avg_acctbal
FROM customer c
JOIN nation n ON c.c_nationkey = n.n_nationkey
GROUP BY n.n_name;
```

- 结果分析：展示不同国家客户的账户余额水平，便于制定分区域信贷策略。
- 运行截图：

```
tpch_db=# SELECT n.n_name AS nation_name, AVG(c.c_acctbal) AS avg_acctbal
FROM customer c
JOIN nation n ON c.c_nationkey = n.n_nationkey
GROUP BY n.n_name;
```

nation_name	avg_acctbal
VIETNAM	4712.1001724137931034
PERU	4301.2785714285714286
ALGERIA	4068.5277049180327869
ARGENTINA	4850.9040677966101695
JAPAN	4962.4638805970149254
IRAQ	4613.1560344827586207
CANADA	4116.1157971014492754
INDONESIA	4971.4110606060606061
FRANCE	3907.3111111111111111
JORDAN	4185.9996296296296296
MOROCCO	5484.4698611111111111
ROMANIA	3941.0362500000000000
IRAN	4206.7551388888888889
EGYPT	4128.4869696969696970
RUSSIA	4321.5410169491525424
SAUDI ARABIA	5495.6919402985074627
BRAZIL	3635.2980882352941176
UNITED STATES	4297.5358333333333333
CHINA	5032.1215517241379310
GERMANY	4280.0992982456140351
INDIA	4566.6970000000000000
UNITED KINGDOM	3828.2900000000000000
MOZAMBIQUE	4584.8072580645161290
KENYA	4901.1100000000000000
ETHIOPIA	3539.6661403508771930
(25 行记录)	

6. 查询没有下过任何订单的客户信息。

- SQL:

```
SELECT c.*
FROM customer c
WHERE NOT EXISTS (
    SELECT 1
    FROM orders o
    WHERE o.o_custkey = c.c_custkey
)
ORDER BY c.c_custkey;

-- Use join
SELECT c.*
FROM customer c
LEFT JOIN orders o ON c.c_custkey = o.o_custkey
WHERE o.o_orderkey IS NULL
ORDER BY c.c_custkey;
```

也是两种方法，把第四题反一下就好。

- 结果分析：LEFT JOIN 搭配空值过滤识别沉默客户，为营销唤醒提供目标客户名单。
- 运行截图：

c_custkey	c_name	c_address	c_nationkey	c_phone	c_acctbal	c_mktsegment	c_comment
3	Customer#000000003	MG9kdT02NBhm	1	11-719-748-3364	7498.12	AUTOMOBILE	deposits eat slyly ironic, even instructions. express foxes detect slyly. b
6	Customer#000000006	sKZz0CsnMD7mp4Xd0YrBvx,LREYKUWAh yVn	20	30-114-968-4951	7638.57	AUTOMOBILE	tions. even deposits boost according to the slyly bold packages. final accou
9	Customer#000000009	xKIAfTjUsCuxfeleNqefumTrJ5	8	18-338-906-3675	8324.07	FURNITURE	r theodolites according to the requests wake thinly excuses: pending request
12	Customer#000000012	9PMKuhzT4Zr1Q	13	23-791-276-1263	3396.49	HOUSEHOLD	to the carefully final braids. blithely regular requests nag. ironic theodo
15	Customer#000000015	YtWggXoDLdwd07b8y,B2aGUQMLJMX1Y,EC,60n	23	33-687-542-7601	2788.52	HOUSEHOLD	platelets. regular deposits detect asymptotes. blithely unusual packages na
18	Customer#000000018	3txG0 AiuFux3zT8Z9NfArnZt	6	16-155-215-1315	5494.43	BUILDING	s sleep. carefully even instructions nag furiously alongside of t
21	Customer#000000021	XYwVpr9YAHdEn	8	18-902-614-8344	1428.25	MACHINERY	quickly final accounts integrate blithely furiously u
24	Customer#000000024	HXAfGfAyjxtdqwint13Y30Z0 4xeLe7U8PqG	13	23-127-851-8031	9255.67	MACHINERY	into beans. fluffily final ideas haggle fluffily
27	Customer#000000027	IS8GIxpBrLpMT0u7	3	13-137-193-2709	5679.84	BUILDING	about the carefully ironic pinto beans. account
30	Customer#000000030	nJDSfELGavU63l8eSNKsKfL8rLJ00kQnYL20JY	1	11-764-165-5076	9321.01	BUILDING	lithely final requests. furiously unusual account
33	Customer#000000033	qFS1M4LueBx0xnn5ib2cswUweg D	17	27-375-391-1200	-78.56	AUTOMOBILE	s. slyly regular accounts are furiously. carefully pending requests
36	Customer#000000036	3TVc2juPzp30,dJ38kWSU	21	31-704-669-5769	4987.27	BUILDING	haggle. enticing, quiet platelets grow quickly bold sheaves. carefully regul
39	Customer#000000039	nnBrg,Pvy33dfkorYE FdeZ60	2	12-387-467-6509	6264.31	AUTOMOBILE	tions. slyly silent excuses slee
42	Customer#000000042	z15rVyyBke	5	15-416-330-4175	8727.01	BUILDING	ssly according to the pinto beans: carefully special requests across the eve
45	Customer#000000045	4v30cpFgo0mMG,CbnF,4mdC	9	19-715-208-9917	9983.38	AUTOMOBILE	nto beans haggle slyly alongside of t
48	Customer#000000048	4UU iPhBupFvenNB	0	10-508-348-5882	3792.50	BUILDING	re fluffily pending foxes. pending, bold platelets sleep slyly. even platele
51	Customer#000000051	uR,wEaiTvod	12	22-344-885-4251	855.87	FURNITURE	eposits. furiously regular requests integrate carefully packages. furious
54	Customer#000000054	,k4vf 5vECGWfy,hostE,	4	14-776-370-4745	868.90	AUTOMOBILE	sual, silent accounts. furiously express accounts cajole special deposits. f
57	Customer#000000057	97XYbsuOPRXPMU	21	31-835-306-1650	4151.93	AUTOMOBILE	ove the carefully special packages. even, unusual deposits sleep slyly pend
60	Customer#000000060	FyodhjwMChsZmUz7J3z0H	12	22-480-575-5866	2741.87	MACHINERY	latelets. blithely unusual courts boost furiously about the packages. blithe
63	Customer#000000063	IXPspVmhZrakII	21	31-952-552-9584	9331.13	AUTOMOBILE	ithely even accounts detect slyly above the fluffily ir
66	Customer#000000066	XbSeQXHIETbJYYtA1A	22	32-213-373-5994	242.77	HOUSEHOLD	le slyly accounts. carefully silent packages benea
69	Customer#000000069	Ltx17n09Mwhtdbe9QZVxNgP98V7xW97uv5H1prEw	9	19-225-978-5670	1709.28	HOUSEHOLD	thely final ideas around the quickly final dependencies affix carefully quic
72	Customer#000000072	putj1mskxE,zs,HqeIA9nqu7dhgHSBVCwDwHcfr	2	12-759-144-9689	-362.86	FURNITURE	ithely final foxes sleep always quickly bold accounts. final wat
75	Customer#000000075	Dh 6Jz,cwxMLKQfRKKIGrzv6pm	18	28-247-803-9025	6684.10	AUTOMOBILE	Instructions cajole even, even deposits. finally bold deposits use above th
78	Customer#000000078	HBDta,ZNagp3U2cSL0kbrfFtkPwzX	9	19-968-700-9191	7136.97	FURNITURE	ests. blithely bold pinto beans h
81	Customer#000000081	SH61PA7J1jncGdnTr	20	30-165-277-3269	2023.71	BUILDING	r packages. fluffily ironic requests cajole fluffily. ironically regular the
84	Customer#000000084	lpz6Fwr9945rnbTmc8PLue1lS1MmAsr CB	11	21-546-818-3802	5174.71	FURNITURE	ly blithe foxes. special asymptotes haggle blithely against the furiously re
87	Customer#000000087	ghGHWSMQL 6JZ6Ev	23	33-869-884-7053	6327.54	FURNITURE	hely ironic requests integrate according to the ironic accounts. slyly regul
90	Customer#000000090	QxC2h7VxxYUWwF7L	16	26-603-491-1238	7354.23	BUILDING	sly across the furiously even
93	Customer#000000093	EHKBrZQdgh	7	17-359-380-5266	2182.52	MACHINERY	press deposits. carefully regular platelets r
96	Customer#000000096	vWL0rmxhRR	8	10-422-845-1202	6323.92	AUTOMOBILE	press requests believe furiously. carefully final instructions snooze carefu
99	Customer#000000099	sZsr01PtCHVS97Lt	15	25-515-237-9232	4088.65	HOUSEHOLD	cajole slyly about the regular theodolites! furiously bold requests nag alon
102	Customer#000000102	UAtfLJ06 fn9zBfKjInkQZlWtqa	19	29-324-978-8538	8462.17	BUILDING	ously regular dependencies nag among the furiously express dinos. blithely f
105	Customer#000000105	4i5JedL SPjg7Kj98Yz3z0B	10	20-793-553-6417	9091.12	MACHINERY	l pains cajole even accounts. quietly final instructi
108	Customer#000000108	GPoeEvk01	5	15-908-619-7526	2259.38	BUILDING	refully ironic deposits sleep. regular, unusual requests wake slyly
111	Customer#000000111	CB5bPy0WRorloj2TBvrK9qp9tHBs	22	32-582-283-7528	6505.26	MACHINERY	ly unusual instructions detect fluffily special deposits-- theodolites nag c
114	Customer#000000114	xAt 5f5AlFIU	14	24-805-212-7646	1027.46	FURNITURE	der the carefully express theodolites are after the packages. packages are.

7. 查询在同一国家的客户对（不重复组合）。

- SQL：

```
SELECT c1.c_name AS customer1, c2.c_name AS customer2, n.n_name AS nation_name
FROM customer c1
JOIN customer c2 ON c1.c_nationkey = c2.c_nationkey AND c1.c_custkey < c2.c_custkey
JOIN nation n ON c1.c_nationkey = n.n_nationkey;
```

- 结果分析：通过自连接并限制主键大小关系避免重复组合，生成同国客户配对清单。
- 运行截图：

customer1	customer2	nation_name
Customer#000000001	Customer#000001428	MOROCCO
Customer#000000001	Customer#000001426	MOROCCO
Customer#000000001	Customer#000001413	MOROCCO
Customer#000000001	Customer#000001383	MOROCCO
Customer#000000001	Customer#000001368	MOROCCO
Customer#000000001	Customer#000001357	MOROCCO
Customer#000000001	Customer#000001320	MOROCCO
Customer#000000001	Customer#000001298	MOROCCO
Customer#000000001	Customer#000001287	MOROCCO
Customer#000000001	Customer#000001256	MOROCCO
Customer#000000001	Customer#000001233	MOROCCO
Customer#000000001	Customer#000001190	MOROCCO
Customer#000000001	Customer#000001188	MOROCCO
Customer#000000001	Customer#000001147	MOROCCO

Customer#000000001	Customer#000001092	MOROCCO
Customer#000000001	Customer#000001090	MOROCCO
Customer#000000001	Customer#000001070	MOROCCO
Customer#000000001	Customer#000001064	MOROCCO
Customer#000000001	Customer#000001029	MOROCCO
Customer#000000001	Customer#000001013	MOROCCO
Customer#000000001	Customer#000000950	MOROCCO
Customer#000000001	Customer#000000945	MOROCCO
Customer#000000001	Customer#000000924	MOROCCO
Customer#000000001	Customer#000000878	MOROCCO
Customer#000000001	Customer#000000856	MOROCCO
Customer#000000001	Customer#000000848	MOROCCO
Customer#000000001	Customer#000000814	MOROCCO
Customer#000000001	Customer#000000811	MOROCCO
Customer#000000001	Customer#000000776	MOROCCO
Customer#000000001	Customer#000000719	MOROCCO
Customer#000000001	Customer#000000711	MOROCCO
Customer#000000001	Customer#000000684	MOROCCO
Customer#000000001	Customer#000000682	MOROCCO
Customer#000000001	Customer#000000668	MOROCCO
Customer#000000001	Customer#000000664	MOROCCO
Customer#000000001	Customer#000000639	MOROCCO
Customer#000000001	Customer#000000627	MOROCCO
Customer#000000001	Customer#000000597	MOROCCO
Customer#000000001	Customer#000000584	MOROCCO
Customer#000000001	Customer#000000570	MOROCCO
Customer#000000001	Customer#000000562	MOROCCO
Customer#000000001	Customer#000000557	MOROCCO
Customer#000000001	Customer#000000555	MOROCCO
Customer#000000001	Customer#000000551	MOROCCO
Customer#000000001	Customer#000000533	MOROCCO
Customer#000000001	Customer#000000532	MOROCCO
Customer#000000001	Customer#000000529	MOROCCO
Customer#000000001	Customer#000000528	MOROCCO
Customer#000000001	Customer#000000515	MOROCCO
Customer#000000001	Customer#000000429	MOROCCO
Customer#000000001	Customer#000000398	MOROCCO
Customer#000000001	Customer#000000395	MOROCCO
Customer#000000001	Customer#000000375	MOROCCO
Customer#000000001	Customer#000000350	MOROCCO
Customer#000000001	Customer#000000325	MOROCCO
Customer#000000001	Customer#000000297	MOROCCO
Customer#000000001	Customer#000000296	MOROCCO
Customer#000000001	Customer#000000267	MOROCCO
Customer#000000001	Customer#000000253	MOROCCO

:|

8. 查询订单总金额超过该客户平均订单金额的订单信息。

- SQL:

```

SELECT o.*
FROM orders o
JOIN (
    SELECT o_custkey, AVG(o_totalprice) AS avg_order_amount
    FROM orders
    GROUP BY o_custkey
) cust_avg ON o.o_custkey = cust_avg.o_custkey
WHERE o.o_totalprice > cust_avg.avg_order_amount;

```

先用子查询查出每个用户的平均订单金额
然后拼到每个用户上再select

• 运行截图：

o_orderkey	o_custkey	o_orderstatus	o_totalprice	o_orderdate	o_orderpriority	o_clerk	o_shippriority	o_comment
1	370	O	172789.49	1996-01-02	5-LOW	Clerk#0000000951	0	nstructions sleep furiously among
3	1234	F	205654.30	1993-10-14	5-LOW	Clerk#0000000955	0	sly final accounts boost, carefully regular ideas cajole carefully, depos
7	302	O	271885.66	1996-01-10	2-HIGH	Clerk#0000000470	0	ly special requests
32	1301	O	198665.57	1995-07-16	2-HIGH	Clerk#0000000616	0	ise blithely bold, regular requests, quickly unusual dep
35	1276	O	194641.93	1995-10-23	4-NOT SPECIFIED	Clerk#0000000259	0	zzle, carefully enticing deposits nag furio
37	862	F	131896.49	1992-06-03	3-MEDIUM	Clerk#0000000456	0	kly regular pinto beans, carefully unusual waters cajole never
39	818	O	326563.37	1996-09-20	3-MEDIUM	Clerk#0000000659	0	ole express, ironic requests: if
67	568	O	182481.16	1996-12-19	4-NOT SPECIFIED	Clerk#0000000547	0	symptotes haggle slyly around the furiously iron
68	286	O	301968.79	1998-04-18	3-MEDIUM	Clerk#0000000440	0	pinto beans sleep carefully, blithely ironic deposits haggle furiously acro
69	845	F	204110.73	1994-06-04	4-NOT SPECIFIED	Clerk#0000000330	0	depths atop the slyly thin deposits detect among the furiously silent accou
71	34	O	260603.38	1998-01-24	4-NOT SPECIFIED	Clerk#0000000271	0	express deposits along the blithely regul
106	1471	O	190970.27	1998-02-20	4-NOT SPECIFIED	Clerk#0000000577	0	hedolites detect slyly alongside of the ent
102	8	O	184806.58	1997-05-09	2-HIGH	Clerk#0000000596	0	slyly according to the asymptotes, carefully final packages integrate furious
103	292	O	118745.16	1996-06-20	4-NOT SPECIFIED	Clerk#0000000090	0	ges, carefully unusual instructions haggle quickly regular f
129	712	F	254281.41	1992-11-19	5-LOW	Clerk#0000000859	0	ing tithes, carefully pending deposits boost about the silently express
138	370	F	140213.54	1992-05-08	2-HIGH	Clerk#0000000036	0	le slyly unusual, regular packages? express deposits det
134	62	F	208281.46	1992-05-01	4-NOT SPECIFIED	Clerk#0000000711	0	lar theodolites boost
135	605	O	230472.84	1995-10-21	4-NOT SPECIFIED	Clerk#0000000804	0	l platelets use according t
163	878	O	177809.13	1997-09-05	3-MEDIUM	Clerk#0000000379	0	y final packages, final foxes since the quickly even
164	8	F	250417.20	1992-10-21	5-LOW	Clerk#0000000209	0	cajole ironic courts, slyly final ideas are slyly, blithely final Tiresias sub
165	274	F	193302.35	1993-01-30	4-NOT SPECIFIED	Clerk#0000000292	0	across the blithely regular accounts, bold
166	1079	O	156207.39	1995-09-12	2-HIGH	Clerk#0000000440	0	lets, ironic, bold asymptotes kindle
192	826	O	194637.57	1997-11-25	5-LOW	Clerk#0000000463	0	y unusual platelets among the final instructions integrate rut
194	619	F	154284.73	1992-04-05	3-MEDIUM	Clerk#0000000352	0	regular requests haggle slyly regular, regular pinto beans, asymptote
195	1355	F	216638.92	1993-12-28	3-MEDIUM	Clerk#0000000216	0	old forges are furiously sheaves, slyly fi
197	326	P	155247.48	1995-04-07	2-HIGH	Clerk#0000000969	0	solve quickly about the even braids, carefully express deposits affix care
198	1103	O	149551.63	1998-01-02	4-NOT SPECIFIED	Clerk#0000000331	0	its, carefully ironic requests sleep, furiously express fox
224	25	F	234050.44	1994-06-18	4-NOT SPECIFIED	Clerk#0000000642	0	r the quickly thin courts, carefully
225	331	P	226028.98	1995-05-25	1-URGENT	Clerk#0000000177	0	s, blithely ironic accounts wake quickly fluffly special acc
226	1276	F	256459.40	1993-03-10	2-HIGH	Clerk#0000000756	0	s are carefully at the blithely ironic acc
229	1118	F	195619.74	1993-12-29	1-URGENT	Clerk#0000000628	0	he fluffly even instructions, furiously i
231	910	F	234383.86	1994-09-29	2-HIGH	Clerk#0000000446	0	packages haggle slyly after the carefully ironic instruct
246	419	F	259466.78	1993-12-29	1-URGENT	Clerk#0000000167	0	dencies, blithely quick packages cajole, ruthlessly final accounts
260	1048	O	268084.69	1996-12-10	3-MEDIUM	Clerk#0000000960	0	lently regular pinto beans sleep after the slyly e
261	461	F	278279.30	1993-06-29	3-MEDIUM	Clerk#0000000310	0	ully fluffly brave instructions, furiousl
262	304	O	173401.63	1995-11-25	4-NOT SPECIFIED	Clerk#0000000551	0	l packages, blithely final pinto beans use carefu
288	71	O	239366.68	1997-02-21	1-URGENT	Clerk#0000000109	0	uriously final requests, even, final ideas det
289	1039	O	174624.55	1997-02-10	3-MEDIUM	Clerk#0000000103	0	slyly, slyly special excuse
322	1336	F	165992.05	1992-03-19	1-URGENT	Clerk#0000000158	0	fully across the slyly bold packages, packages against the quickly regular i
323	392	F	121127.17	1994-03-26	1-URGENT	Clerk#0000000959	0	arefully pending foxes sleep blithely, slyly express account
326	760	O	325448.68	1995-06-04	2-HIGH	Clerk#0000000466	0	requests, furiously ironic asymptotes mold carefully alongside of the blit
353	19	F	224983.69	1993-12-31	5-LOW	Clerk#0000000449	0	quiet ideas sleep, even instructions cajole slyly, silently spe
354	1384	O	231311.22	1996-03-14	2-HIGH	Clerk#0000000511	0	ly regular ideas wake across the slyly silent ideas, final deposits eat b
356	1469	F	189160.02	1994-06-30	4-NOT SPECIFIED	Clerk#0000000944	0	as wake along the bold accounts, even,
357	604	O	138936.83	1996-10-09	2-HIGH	Clerk#0000000301	0	e blithely about the express, final accounts, quickl
358	23	F	362024.17	1993-09-20	2-HIGH	Clerk#0000000392	0	l, silent instructions are slyly, silently even de
359	776	F	214770.97	1994-12-19	3-MEDIUM	Clerk#0000000934	0	n dolphins, special courts above the carefully ironic requests use
384	1132	F	191725.12	1992-03-03	5-LOW	Clerk#0000000206	0	, even accounts use furiously packages, slyly ironic pla
387	24	O	197820.44	1997-01-26	4-NOT SPECIFIED	Clerk#0000000168	0	are carefully among the quickly even deposits, furiously silent req
388	448	F	161560.04	1992-12-16	4-NOT SPECIFIED	Clerk#0000000356	0	ar foxes above the furiously ironic deposits nag slyly final requ
390	1027	O	232256.36	1998-04-07	5-LOW	Clerk#0000000404	0	xpress asymptotes use among the regular, final pinto b
419	1163	O	159079.22	1996-10-01	3-MEDIUM	Clerk#0000000376	0	osits, blithely pending theodolites boost carefully
428	902	O	269064.47	1995-10-31	4-NOT SPECIFIED	Clerk#0000000756	0	leep carefully final excuses, fluffly pending requests unwind carefully above
429	733	O	155333.71	1997-05-31	4-NOT SPECIFIED	Clerk#0000000049	0	apple carefully across the accounts, regular accounts eat fluffl
448	1498	O	157247.56	1995-08-21	3-MEDIUM	Clerk#0000000597	0	regular, express foxes use blithely, quic
450	475	P	213638.07	1995-03-05	4-NOT SPECIFIED	Clerk#0000000293	0	d theodolites, boldly bold foxes since the pack
453	442	O	343004.49	1997-05-26	5-LOW	Clerk#0000000504	0	ss foxes, furiously regular ideas sleep according to t
455	121	O	190711.32	1996-12-84	1-URGENT	Clerk#0000000796	0	about the final platelets, dependen
481	304	F	160370.14	1992-10-08	2-HIGH	Clerk#0000000230	0	ly final ideas, packages haggle fluffly

9. 查询供应了所有类型商品的供应商信息。

• SQL:

```

SELECT s.*
FROM supplier s
WHERE NOT EXISTS (
    -- 找出供应商s没有供应的零件
    SELECT p.p_type
    FROM part p
    -- 不存在找不到该记录
    WHERE NOT EXISTS (
        -- 找出 partsupp 供应关系表中, 属于供应商 s, 对应零件 p 的那条供应记录。
        -- 即确认供应商 s 是否供应了零件 p
        SELECT 1
        FROM partsupp ps

```

```

        WHERE ps.ps_suppkey = s.s_suppkey AND ps.ps_partkey = p.p_partkey
    )
);

-- 使用HAVING
SELECT s.*
-- 直接全拼到供应商表上
-- 看供应商表提供的独特商品p个数和总表个数一样不一样就可以了
FROM supplier s
JOIN partsupp ps ON s.s_suppkey = ps.ps_suppkey
JOIN part p ON ps.ps_partkey = p.p_partkey
GROUP BY s.s_suppkey, s.s_name, s.s_address, s.s_nationkey, s.s_phone, s.s_acctbal,
s.s_comment
HAVING COUNT(DISTINCT p.p_type) = (SELECT COUNT(DISTINCT p_type) FROM part);

```

两个 NOT EXISTS 这个逻辑确实有点绕，难想出来...

具体解释看注释

值得一提的是这里没有供应商供应了所有商品。

- 运行截图：

```

tpch_db=# SELECT s.*
FROM supplier s
WHERE NOT EXISTS (
    SELECT p.p_type
    FROM part p
    WHERE NOT EXISTS (
        SELECT 1
        FROM partsupp ps
        WHERE ps.ps_suppkey = s.s_suppkey
        AND ps.ps_partkey = p.p_partkey
    )
);
 s_suppkey | s_name | s_address | s_nationkey | s_phone | s_acctbal | s_comment
-----+-----+-----+-----+-----+-----+-----
(0 行记录)

tpch_db=# SELECT s.*
FROM supplier s
JOIN partsupp ps ON s.s_suppkey = ps.ps_suppkey
JOIN part p ON ps.ps_partkey = p.p_partkey
GROUP BY s.s_suppkey, s.s_name, s.s_address, s.s_nationkey, s.s_phone, s.s_acctbal, s.s_comment
HAVING COUNT(DISTINCT p.p_type) = (SELECT COUNT(DISTINCT p_type) FROM part);
 s_suppkey | s_name | s_address | s_nationkey | s_phone | s_acctbal | s_comment
-----+-----+-----+-----+-----+-----+-----
(0 行记录)

```

10. 查询总价大于所处月份平均总价的订单。

- SQL:

```

-- Solution1:
-- 遍历 orders 表的每一行，对于每一行：
-- 查看该订单的年份和月份。
-- 找到所有与它同年同月的其他订单（即它所在的分区）。
-- 计算这个分区内所有订单的 o_totalprice 的平均值。
-- 将这个算出的平均值作为新的一列 avg_monthly_price 附加到当前行上。
WITH MonthlyAvg AS (
    SELECT

```

```

*,
-- 2. 添加一个计算出的新列
-- 关键字 OVER 表明这是一个窗口函数, 它不会像 GROUP BY 那样把多行压缩成一行, 而是在保持原有行数不变的情况下, 为每一行计算一个附加值。
    AVG(o_totalprice) OVER (PARTITION BY EXTRACT(YEAR FROM o_orderdate), EXTRACT(MONTH FROM o_orderdate)) AS avg_monthly_price
FROM
    orders
)
SELECT MonthlyAvg.o_orderkey, o_totalprice, avg_monthly_price
FROM MonthlyAvg
WHERE o_totalprice > avg_monthly_price
ORDER BY o_orderkey;

-- Solution2:
SELECT o.*
FROM orders o
JOIN (
    -- 求每个年/月的平均总价
    SELECT
        EXTRACT(YEAR FROM o_orderdate) AS order_year,
        EXTRACT(MONTH FROM o_orderdate) AS order_month,
        AVG(o_totalprice) AS monthly_avg
    FROM orders
    GROUP BY EXTRACT(YEAR FROM o_orderdate), EXTRACT(MONTH FROM o_orderdate)
) monthly_avg ON EXTRACT(YEAR FROM o.o_orderdate) = monthly_avg.order_year
    AND EXTRACT(MONTH FROM o.o_orderdate) = monthly_avg.order_month
WHERE o.o_totalprice > monthly_avg.monthly_avg;

```

AI直接上了WITH这个没学过的语法...

第二种虽然看着烦但是容易理解一点

- 运行截图:

o_orderkey	o_custkey	o_orderstatus	o_totalprice	o_orderdate	o_orderpriority	o_clerk	o_shippriority	o_comment
1	370	D	172799.49	1996-01-02	5-LOW	Clerk#000000951	0	Instructions sleep furiously among
3	1234	F	205654.30	1993-10-14	5-LOW	Clerk#000000955	0	sly final accounts boost, carefully regular ideas cajole carefully, depas
7	392	O	21885.66	1996-01-10	2-HIGH	Clerk#000000470	0	ly special requests
32	1301	O	198665.57	1995-07-16	2-HIGH	Clerk#000000616	0	ise blithely bold, regular requests, quickly unusual dep
35	1276	O	194641.93	1995-10-23	4-NOT SPECIFIED	Clerk#000000259	0	zzle, carefully enticing deposits nag furio
39	818	O	326565.37	1996-09-20	3-MEDIUM	Clerk#000000659	0	ole express, ironic requests: ir
67	568	O	182481.16	1996-12-19	4-NOT SPECIFIED	Clerk#000000547	0	symptotes haggle slyly around the furiously iron
68	286	O	301968.79	1998-04-18	3-MEDIUM	Clerk#000000440	0	pinto beans sleep carefully, blithely ironic deposits haggle furiously acro
69	845	F	204110.73	1994-06-04	4-NOT SPECIFIED	Clerk#000000330	0	depths atop the slyly thin deposits detect among the furiously silent accou
71	34	O	260603.38	1990-01-24	4-NOT SPECIFIED	Clerk#000000271	0	express deposits along the blithely regul
100	1471	O	198978.27	1998-02-28	4-NOT SPECIFIED	Clerk#000000577	0	theadolites detect slyly alongside of the ent
102	8	O	184806.58	1997-05-09	2-HIGH	Clerk#000000596	0	slyly according to the asymptotes, carefully final packages integrate furious
129	712	F	254281.41	1992-11-19	5-LOW	Clerk#000000859	0	ing tithes, carefully pending deposits boost about the silently express
130	370	F	140213.54	1992-05-08	2-HIGH	Clerk#000000036	0	le slyly unusual, regular packages? express deposits det
131	928	F	140726.47	1994-06-08	3-MEDIUM	Clerk#000000025	0	after the fluffily special foxes integrate s
134	62	F	208201.46	1992-05-01	4-NOT SPECIFIED	Clerk#000000711	0	lar theodolites boost
135	605	O	230472.84	1995-10-21	4-NOT SPECIFIED	Clerk#000000084	0	l platelets use according t
163	878	O	177809.13	1997-09-05	3-MEDIUM	Clerk#000000379	0	y final packages, final foxes since the quickly even
164	8	F	250417.20	1992-10-21	5-LOW	Clerk#000000209	0	cajole ironic courts, slyly final ideas are slyly, blithely final Tiresias sub
165	274	F	193302.35	1993-01-30	4-NOT SPECIFIED	Clerk#000000292	0	across the blithely regular accounts, bold
166	1079	O	158207.39	1995-09-12	2-HIGH	Clerk#000000440	0	lets, ironic, bold asymptotes kindle
192	826	O	194637.57	1997-11-25	5-LOW	Clerk#000000483	0	y unusual platelets among the final instructions integrate rut
194	619	F	154284.73	1992-04-05	3-MEDIUM	Clerk#000000044	0	regular requests haggle slyly regular, regular pinto beans, asymptote
195	1355	F	216638.92	1993-12-28	3-MEDIUM	Clerk#000000216	0	old forges are furiously sheaves, slyly fi
197	326	P	155247.48	1995-04-07	2-HIGH	Clerk#000000969	0	solve quickly about the even braids, carefully express deposits affix care
198	1103	O	140551.63	1990-01-02	4-NOT SPECIFIED	Clerk#000000331	0	its, carefully ironic requests sleep, furiously express fox
224	25	F	234050.44	1994-06-18	4-NOT SPECIFIED	Clerk#000000042	0	r the quickly thin courts, carefully
225	331	P	226028.98	1995-05-25	1-URGENT	Clerk#000000177	0	s, blithely ironic accounts wake quickly fluffily special acc
226	1276	F	256459.40	1993-03-10	2-HIGH	Clerk#000000756	0	s are carefully at the blithely ironic acc
229	1118	F	195619.74	1993-12-29	1-URGENT	Clerk#000000628	0	he fluffily even instructions, furiously i
231	910	F	234383.86	1994-09-29	2-HIGH	Clerk#000000446	0	packages haggle slyly after the carefully ironic instruct
258	419	F	250466.78	1993-12-29	1-URGENT	Clerk#000000167	0	encies, blithely quick packages cajole, ruthlessly final accounts
260	1048	O	260084.69	1996-12-10	3-MEDIUM	Clerk#000000960	0	lently regular pinto beans sleep after the slyly e
261	461	F	278279.30	1993-06-29	3-MEDIUM	Clerk#000000310	0	ully fluffily brave instructions, furiouslyl
262	304	O	173401.63	1995-11-25	4-NOT SPECIFIED	Clerk#000000551	0	l packages, blithely final pinto beans use carefu
268	71	O	230366.68	1997-02-21	1-URGENT	Clerk#000000109	0	uriously final requests, even, final ideas det
280	1039	O	174624.55	1997-02-10	3-MEDIUM	Clerk#000000103	0	sily, slyly special excuse
322	1336	F	165992.05	1992-03-19	1-URGENT	Clerk#000000158	0	fully across the slyly bold packages, packages against the quickly regular i
326	760	O	325448.68	1995-06-04	2-HIGH	Clerk#000000466	0	requests, furiously ironic asymptotes mold carefully alongside of the blit
353	19	F	224983.69	1993-12-31	5-LOW	Clerk#000000449	0	quiet ideas sleep, even instructions cajole slyly, silently spe
354	1304	O	231311.22	1996-03-14	2-HIGH	Clerk#000000511	0	ly regular ideas wake across the slyly silent ideas, final deposits eat b
356	1469	F	189160.02	1994-06-30	4-NOT SPECIFIED	Clerk#000000944	0	as wake along the bold accounts, even,
357	604	O	138936.83	1996-10-09	2-HIGH	Clerk#000000301	0	e blithely about the express, final accounts, quickl
358	23	F	362024.17	1993-09-20	2-HIGH	Clerk#000000392	0	l, silent instructions are slyly, silently even de
359	776	F	214770.97	1994-12-19	3-MEDIUM	Clerk#000000934	0	n dolphins, special courts above the carefully ironic requests use
384	1132	F	191275.12	1992-03-03	5-LOW	Clerk#000000208	0	, even accounts use furiously packages, slyly ironic pla
387	34	O	197839.44	1997-01-26	4-NOT SPECIFIED	Clerk#000000768	0	are carefully among the quickly even deposits, furiously silent req
388	448	F	161560.04	1992-12-16	4-NOT SPECIFIED	Clerk#000000356	0	ar foxes above the furiously ironic deposits nag slyly final requ
390	1027	O	232256.36	1998-04-07	5-LOW	Clerk#000000404	0	xpress asymptotes use among the regular, final pinto b
419	1163	O	159079.22	1996-10-01	3-MEDIUM	Clerk#000000376	0	osits, blithely pending theodolites boost carefully
420	902	O	269064.47	1995-10-31	4-NOT SPECIFIED	Clerk#000000756	0	leep carefully final excuses, fluffily pending requests unwind carefully above
422	731	O	155533.71	1997-05-31	4-NOT SPECIFIED	Clerk#000000049	0	apple carefully across the accounts, regular accounts eat fluffi
448	1498	O	157247.56	1995-08-21	3-MEDIUM	Clerk#000000597	0	regular, express foxes use blithely, quic
450	475	P	213638.07	1995-03-05	4-NOT SPECIFIED	Clerk#000000293	0	d theodolites, boldly bold foxes since the pack
453	442	O	343004.49	1997-05-26	5-LOW	Clerk#000000504	0	ss foxes, furiously regular ideas sleep according to t
455	121	O	190711.32	1996-12-04	1-URGENT	Clerk#000000796	0	about the final platelets, dependen
481	304	F	160370.14	1992-10-08	2-HIGH	Clerk#000000230	0	ly final ideas, packages haggle fluffily
482	1252	O	197194.23	1996-03-26	1-URGENT	Clerk#000000295	0	ts, deposits wake: final acco
484	544	O	331553.32	1997-01-03	3-MEDIUM	Clerk#000000545	0	grouches use, furiously bold accounts maintain, bold, regular deposits

3. 实验总结

本实验系统梳理了 TPC-H 数据集上的单表与多表查询场景，熟练运用了分组、连接与窗口函数等 SQL 方法。通过逐条验证运行结果，进一步理解了筛选条件与数据特征之间的关系，为后续复杂查询与性能调优奠定了实践基础。

总的来说，还是需要练一练的，尤其是多表查询还是有很难的自己没法写出来的题目，需要寻求AI的帮助...